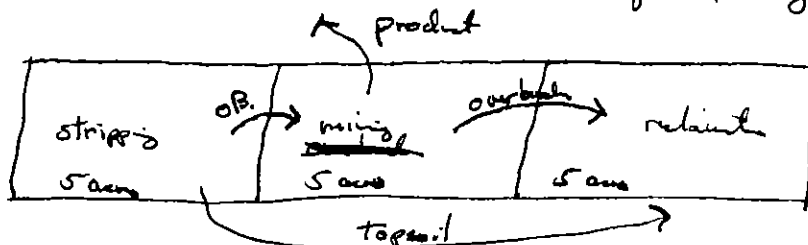




- 1) Martin Marietta Cement - Larimer County Quarry
located in Larimer Co., 7 miles SW of Berthoud
Elevation 5375'
- 2) Surface & Mineral Rights owned by Martin Marietta Corp
- 3) Local Gov't Approval - obtained June 15, 1972 from the
Larimer Co. Commission.
- 4) Size - of permit area is 196 acres
- 5) General Description - The area to be mined is
used primarily for dryland pasture. The area generally
slopes from the NW to the SE. The pits are
located on outcrops where the ore body is at least
partially exposed. The operation will be eliminating the
ridges created by these outcrops. There are to be
4 pits - 2 high grade limestone - 2 low grade limestone
No surface water or shallow groundwater occurs on the
site, though irrigation water is available to the property.
Overburden in the ~~one~~ pit areas varies from 0 to 25
in depth, of that amount up to 8" can be topsoil,
and 10' can be clays resulting from ~~under~~ the
shales which overlie the ore. Vegetation is described as a
poor range
- 6) Mining Plan -
The operation is divided into 4 Pits;
pits A and B are high grade, pits C & D are low grade.
~~Currently pit A is being operated,~~
~~Pit A - is currently being mined; mining started in~~
~~the NE corner of the pit and is proceeding to the SW~~

(next page)

At any given time only 15 acres should be in some stage of development, mining, and reclamation. The mining plan calls for one 5 acre parcel to be mined out and backfilled with ~~material~~ overburden from the second 5 acre parcel. While the second 5 acre parcel is being mined topsoil from the 3rd 5 acres will be stripped ~~from the~~ and spread on the first 5 acres and then 1st 5 acres can be revegetated. The attempt is to minimize disturbance ~~at~~ any one time to 15 acres, with 5 acres each in development, mining, and reclamation.



1973-2003 Pit A - is currently being mined. The operation started in the NE corner of the pit and mining is progressing ~~in~~ to the SW. Again only 15 acres is to be disturbed at any given time.

Each 5 acre section will be mined perpendicular to the strike of the orebody. Each year 5 acres is opened up. After Pit A is mined out the operation will proceed to Pit B. Pit B operation is the same except that mining will proceed from the south to the north.

Pit B →
1995-2023

Pit D →
2039-2080

After Pit B, pit D will be mined, just like the others and in a South to north direction.

Pit C →
2013-2045

Pit C's western portion will be mined ~~at~~ after Pit D. If material is still needed by Martin Marietta after the Western portion is mined the Eastern portion will be mined.

Reclamation Plan -

All areas will be returned to dryland pasture. The overburden will be topped with 6" to 10" of topsoil and sealed with 25-25 lbs of the seed mixture and fertilized with 200-300 lbs of ammonium nitrate. The seed mixture is

Dryland Mix

39% winter rye
 13% range alfalfa
 9% smooth bromegrass
 8% ~~intermediate~~ intermediate wheatgrass
 7% slender wheatgrass
 8% Nordan crested wheatgrass
 6% Russian wild rye
 10% Jose tall wheatgrass

Irrigation water via a sprinkler system is available to initiate growth.

Vegetation Soils Water Wildlife

Water no problems

Soils no problems

Vegetation is described as overgrazed, declining condition 25% cover.

Wildlife had Div of Wildlife very happy and they concur that no long term vegetation effects will occur.

Bond

major - 15 x 1500.00 = 22,500

minor 10 x 300 = 3000
 25,500

10%
 + 2,500
\$ 28,000

Marta Pericetti

Larimer County Quarry

77-341

Water Resources says: Lyons Quarry *

water rights for irrigated

reclaimed land presented; water required for reclamation
not presented

- water for dust suppression not stated nor source

- lakes resulting mean 300 acre/ft annual loss from
over appropriated watershed

- no evidence offered to demonstrate non-injury to
vested water rights.

- Water Resources recommends no approval?

A - legal $\frac{445.30}{19.84} \rightarrow 543.14$ affluents approx. 151 Sec 35-
45 Sec 36
196

B ok.

C quit

D ok

E depth based on drill should be used.
dryland species mix check ↑

F check reclamation contour against previous map

G - no known aquifers no surface water no impoundments
420000 gallons monthly for dust/reclamation used.

H - Wildlife ok

I - ok

J - ok

K - ok

L - confidential

M - ok

N - ok

O - ok